

Please write clearly in block capitals.

Centre number 

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Forename(s) \_\_\_\_\_

Candidate signature \_\_\_\_\_

I declare this is my own work.

# INTERNATIONAL GCSE COMPUTER SCIENCE

## Paper 2 Concepts and principles of computer science

Wednesday 28 May 2025

07:00 GMT

Time allowed: 2 hours

### Materials

You will need no other materials.

### Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of the page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this booklet. Cross through any work you do not want to be marked.
- You are not allowed to use a calculator.

### Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 80.
- You may use a bilingual dictionary.
- You may **not** use an English dictionary.

| For Examiner's Use |      |
|--------------------|------|
| Question           | Mark |
| 1                  |      |
| 2                  |      |
| 3                  |      |
| 4                  |      |
| 5                  |      |
| 6                  |      |
| 7                  |      |
| 8                  |      |
| 9                  |      |
| 10                 |      |
| 11                 |      |
| 12                 |      |
| 13                 |      |
| 14                 |      |
| <b>TOTAL</b>       |      |



Answer **all** questions in the spaces provided.

**0 1 . 1**

Convert the binary number 01011001 to decimal.

**[1 mark]**

\_\_\_\_\_

\_\_\_\_\_

Answer \_\_\_\_\_

**0 1 . 2**

Convert the hexadecimal number 3C to binary.

**[2 marks]**

\_\_\_\_\_

\_\_\_\_\_

Answer \_\_\_\_\_

**0 1 . 3**

Add together the two binary numbers shown in **Figure 1**.

Give your answer in binary.

**Figure 1**

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 0 |
| + | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 |

Answer

\_\_\_\_\_

\_\_\_\_\_

**[2 marks]**



0 1 . 4

Tick (✓) **one** box to show which binary shift could be used to divide a number by 8**[1 mark]**

|          | Shift                       | Tick one |
|----------|-----------------------------|----------|
| <b>A</b> | Left shift of two places    |          |
| <b>B</b> | Left shift of three places  |          |
| <b>C</b> | Left shift of four places   |          |
| <b>D</b> | Right shift of two places   |          |
| <b>E</b> | Right shift of three places |          |
| <b>F</b> | Right shift of four places  |          |

6

0 2 . 1

Define the term abstraction.

**[1 mark]**


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0 2 . 2

Define the term decomposition.

**[2 marks]**


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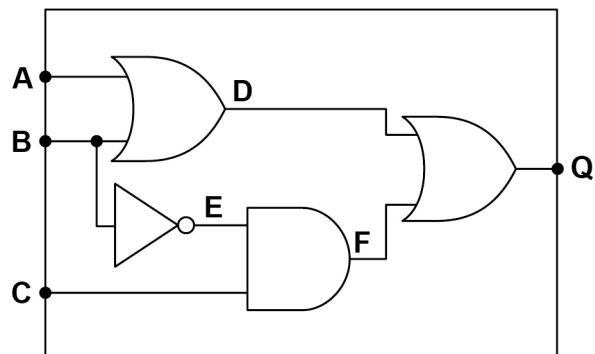
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3

**Turn over for the next question****Turn over ►**

0 3 . 1

**Figure 2** shows a logic circuit with three inputs **A**, **B** and **C** and one output **Q**. The values **D**, **E** and **F** represent the outputs of three of the gates in the circuit.

**Figure 2**

Complete the truth table for the circuit in **Figure 2**.

**[3 marks]**

| Inputs |   |   |   |   |   |  | Output |
|--------|---|---|---|---|---|--|--------|
| A      | B | C | D | E | F |  | Q      |
| 0      | 0 | 0 |   |   |   |  |        |
| 0      | 0 | 1 |   |   |   |  |        |
| 0      | 1 | 0 |   |   |   |  |        |
| 1      | 1 | 1 |   |   |   |  |        |



|   |   |   |   |
|---|---|---|---|
| 0 | 3 | . | 2 |
|---|---|---|---|

Draw a logic circuit to represent the Boolean equation:

$$Q = A \cdot \overline{B} + C$$

**[3 marks]****Turn over for the next question**

|   |
|---|
| 6 |
|---|

**Turn over ►**

|   |   |
|---|---|
| 0 | 4 |
|---|---|

Two components of a CPU are the arithmetic logic unit (ALU) and the clock.

|   |   |   |   |
|---|---|---|---|
| 0 | 4 | . | 1 |
|---|---|---|---|

Name **one** other component of the CPU.

[1 mark]

---

|   |   |   |   |
|---|---|---|---|
| 0 | 4 | . | 2 |
|---|---|---|---|

State **one** example of a task that would be carried out by the ALU.

[1 mark]

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|   |   |   |   |
|---|---|---|---|
| 0 | 4 | . | 3 |
|---|---|---|---|

Describe the purpose of the clock.

[1 mark]

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|   |
|---|
| — |
| 3 |



0 5

ROM and RAM are types of main memory.

0 5 . 1

Describe what main memory is.

[1 mark]

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0 5 . 2

Describe **one** difference between ROM and RAM.

[1 mark]

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0 5 . 3

A computer has 16 gigabytes of RAM.

Calculate how many bytes of RAM the computer has.

[1 mark]

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Answer \_\_\_\_\_ bytes

3

Turn over for the next question

Turn over ►

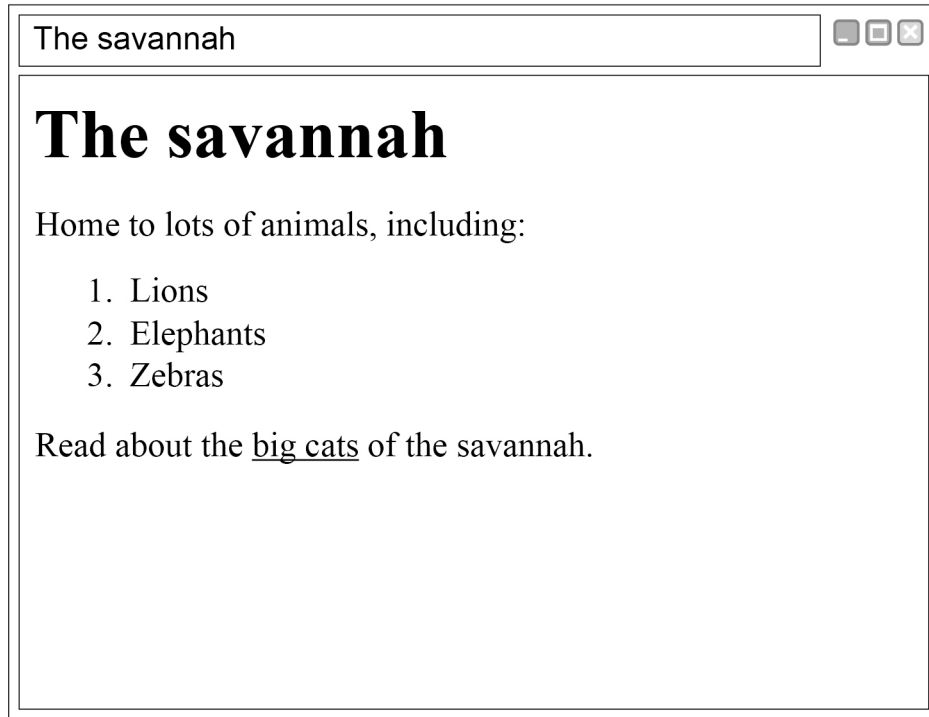


**0 6**

A student is making a website.

Their plan for the first page is shown in **Figure 3**.

**Figure 3**



When the text "big cats" is clicked on, the page `cats.html` will be displayed.





**0 6 . 1**Complete the HTML below to create the page shown in **Figure 3**.**[5 marks]**

```
<html>
  <head>
    <title>The savannah</title>
  </head>

  <body>
```

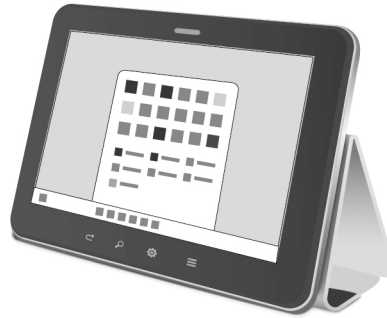
```
    </body>
  </html>
```

**Question 6 continues on the next page****Turn over ►**



A company is designing a new tablet computer. **Figure 4** shows the tablet computer.

**Figure 4**



0 7 . 2

State **two** reasons why a solid state drive (SSD) would be a better choice of secondary storage device for the tablet computer than a magnetic hard disk drive.

**[2 marks]**

Reason 1 \_\_\_\_\_  
\_\_\_\_\_

Reason 2 \_\_\_\_\_  
\_\_\_\_\_

0 7 . 3

State **two** benefits of providing access to cloud storage on a tablet computer.

**[2 marks]**

Benefit 1 \_\_\_\_\_  
\_\_\_\_\_

Benefit 2 \_\_\_\_\_  
\_\_\_\_\_

8

**Turn over for the next question**

**Turn over ►**



**0 8 . 1**

Email confirmations are a cyber security measure.

Describe the purpose of email confirmations.

**[1 mark]**

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**0 8 . 2**Describe **one** limitation of the use of email confirmations.**[1 mark]**

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**0 8 . 3**

Explain how automatic software updates can improve the security of a computer system.

**[2 marks]**

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**4**

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**0 9**

A relational database stores data about rooms and room bookings at a venue.

There are two tables in the database: **Room** and **Booking**.

**Figure 5** shows the data in the two tables.

**Figure 5**

**Room**

| RoomID | MaxCapacity | Cost | Floor | SoundSystem |
|--------|-------------|------|-------|-------------|
| 1      | 20          | 500  | 0     | True        |
| 2      | 20          | 510  | 2     | True        |
| 3      | 22          | 450  | 1     | False       |
| 4      | 30          | 800  | 2     | True        |
| 5      | 10          | 150  | 1     | True        |
| 6      | 50          | 1200 | 3     | True        |

**Booking**

| BookingID | RoomID | Day     | Contact |
|-----------|--------|---------|---------|
| 1         | 4      | Monday  | AKE     |
| 2         | 4      | Tuesday | DBA     |
| 3         | 5      | Tuesday | DBA     |
| 4         | 6      | Friday  | DCO     |
| 5         | 6      | Monday  | BRK     |
| 6         | 1      | Monday  | DBA     |
| 7         | 5      | Friday  | MMF     |

**0 9****1**

Define the term relational database.

**[2 marks]**

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**0 9****2**

State the most suitable data type for the **SoundSystem** field in the **Room** table.

**[1 mark]**

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**0 9****3**

Explain why the **RoomID** field in the **Booking** table cannot be used as the primary key.

**[1 mark]**

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**Figure 6** shows an SQL query.

**Figure 6**

```
SELECT BookingID, Room.RoomID, Cost
FROM Room, Booking
WHERE Room.RoomID = Booking.RoomID
AND Floor > 1
AND (Day = 'Monday' OR Day = 'Tuesday')
```

0 9 . 4

List the records that will be returned when the query in **Figure 6** is executed on the relational database in **Figure 5**.

**[4 marks]**

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**Figure 7** shows an SQL command. This query should remove all the bookings made by the contact DBA. There are errors in the command.

**Figure 7**

```
REMOVE FROM Booking
IF Contact = 'DBA'
```

0 9 . 5

Write a corrected version of the command in **Figure 7**.

**[2 marks]**

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10

**Turn over for the next question**

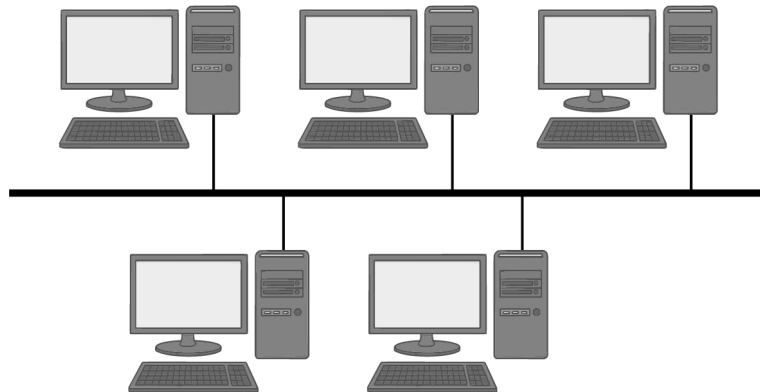
**Turn over ►**



1 0

Figure 8 shows a diagram of a local area network.

**Figure 8**



1 0 . 1

State the topology of the network in **Figure 8**.

[1 mark]

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1 0 . 2

State **one** reason why a business might decide to use the network topology in **Figure 8**.

[1 mark]

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1 0 . 3

State **one** reason why a business might decide not to use the network topology in **Figure 8**.

[1 mark]

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**[4 marks]**

[illegible]

**[1 mark]**

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**[2 marks]**

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**10**



1 1 . 1

State **two** reasons why the digital recording of an analogue sound will not be the same quality as the analogue sound.

**[2 marks]**


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1 1 . 2

A sound that lasts for 3 minutes is recorded using:

- a sample rate of 2000 Hz
- a sample resolution of 2 bytes.

The recording is then stored in a file.

Calculate the file size in kilobytes.

**[3 marks]**


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Answer \_\_\_\_\_ kilobytes

5



**1 2 . 1** Exactly four characters are represented in a character set.

State the minimum number of bits needed to represent any character in this character set.

**[1 mark]**

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**1 2 . 2** The ASCII character set can represent 128 characters.

The ASCII code for the character M is 77

State the bit pattern for the ASCII code for the character P

**[1 mark]**

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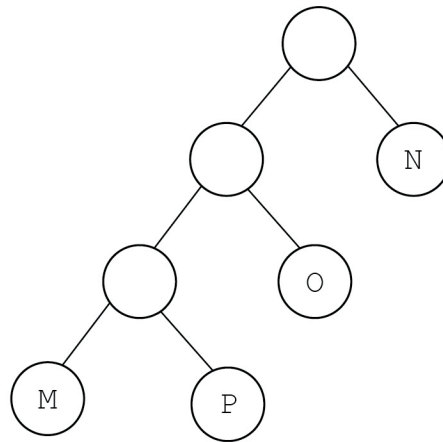
**Question 12 continues on the next page**

**Turn over ►**



**Figure 9** shows a Huffman tree. Using the Huffman code produced from this tree the character **N** is represented by the bit pattern 1

**Figure 9**



1 2 3

What string is represented by the bit pattern 00001001 if the Huffman code from **Figure 9** has been used?

[1 mark]

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1 2 4

The string NOOMOPON would use 56 bits if represented using ASCII.

How many fewer bytes would the string NOOMOPON use if represented using the Huffman code from **Figure 9**?

[2 marks]

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Answer \_\_\_\_\_ bytes



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**1 3 . 1**

Tick (✓) **one** box to show which of the following can be used to write a program that does **not** need to be translated before it can be executed.

**[1 mark]**

|          |                     | Tick one |
|----------|---------------------|----------|
| <b>A</b> | Assembly language   |          |
| <b>B</b> | High-level language |          |
| <b>C</b> | Machine code        |          |
| <b>D</b> | SQL                 |          |

|   |   |   |   |
|---|---|---|---|
| 1 | 3 | . | 2 |
|---|---|---|---|

Describe **three** differences between the way compilers and interpreters operate.

**[3 marks]**

[illegible]

4

**Turn over for the next question**

**Turn over ►**



- 1 4 . 1** A bubble sort is being used to sort a list of names into alphabetical order.

Complete the four blank rows in the table to show the results of each pass through the list.

**[4 marks]**

| Jada | Zuri | Nial | Nico | Chev | Cass |
|------|------|------|------|------|------|
|      |      |      |      |      |      |
|      |      |      |      |      |      |
|      |      |      |      |      |      |
|      |      |      |      |      |      |
| Cass | Chev | Jada | Nial | Nico | Zuri |

- 1 4 . 2** State the maximum size of list that is guaranteed to be fully sorted after four passes of a bubble sort.

**[1 mark]**

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- 1 4 . 3** A merge sort could be used instead of a bubble sort.

State **one** advantage of using a merge sort instead of a bubble sort.

**[1 mark]**

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**END OF QUESTIONS**



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2 8



2 5 6 Y 9 2 1 0 / 2

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